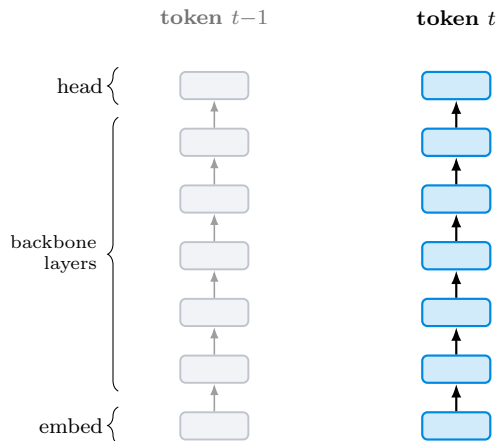


Inference-time information flows

Depth recurrence OFF

Vanilla transformer

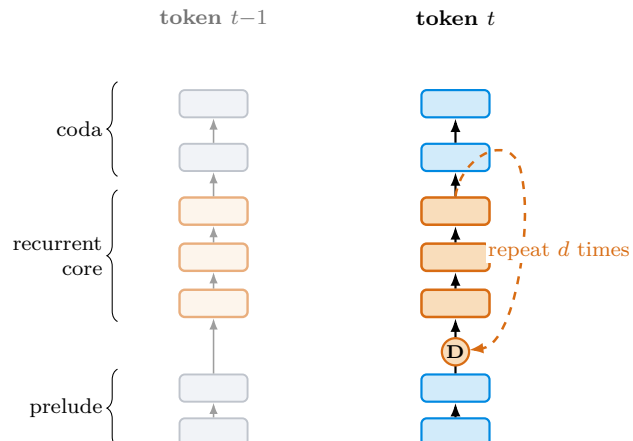
Each token is processed once; there is no temporal state and no depth loop.



Depth recurrence ON

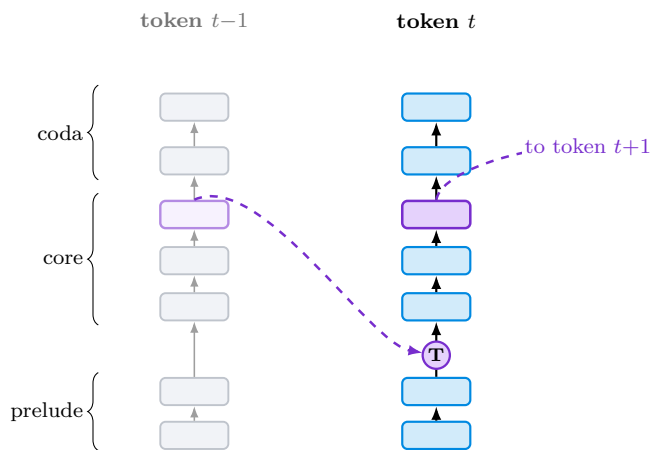
Depth-recurrent / looped

Orange layers form the recurrent core; its top-layer output is reused within token t .



Temporally recurrent

Temporal state comes from the top core layer of token $t-1$ and is read by token t .



Depth + temporal hybrid

Orange marks recurrent-core layers; purple marks the T-source layer.

